



Colorado River Basin Mid-Term Modeling System (CRMMS) Operational Reference

24-Month Study Mode

1. Receiving the CRMMS Model

The CRMMS model initialized in 24-Month Study mode will be available monthly. Mexico modelers can access the model using their secure FTP site. U.S. modelers can receive the model by email (encrypted zip file) upon request.

2. 24-Month Study Mode Set Up

The CRMMS model file provided will be in 24-Month Study Mode and initialized with all input data, output data, and model run configuration set identical to the published 24-Month Study. The ruleset is saved with the model file, so the ruleset will load upon opening the model file.

The 24-Month Study has a run duration that spans 24 to 32 months, depending on the start month of the run. Model runs performed in February through September have a run duration that end in September of the last out-year (i.e. end of water year, see Figure 1).

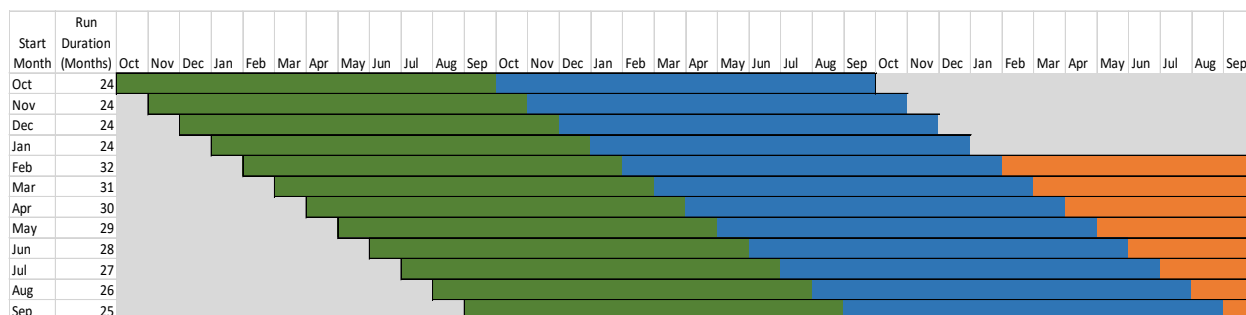


Figure 1: The model run duration of the 24-Month Study. The green bar represents the first 12 months of the model run and the blue bar represents the second 12 months. The orange bar at the end shows the remaining months of the model run through the end of the water year.

3. Running 24-Month Study Mode

The following instructions will result in duplicating the published 24-Month Studies. This is a single deterministic model run being simulated with the hydrology and water use assumptions that are currently loaded in the model (i.e., Most Probable inflow scenario).

- a. Open the model (the ruleset will load automatically).
- b. If you are starting with the CRMMS model in 24-Month Study mode, no additional actions are required (i.e., skip step c below).
- c. If you are starting with the CRMMS model in Ensemble mode, go to the Scripts menu and select Script **3_Change Model to 24MS Mode**.

***Note** – If you have run the model in Ensemble Mode, the outflows from all reservoirs will NOT match the official published 24-Month Study! Re-open the model provided in 24-Month Study mode to ensure the outflows match the official published 24-Month Study. Do not run any scripts once the model has been re-opened in 24-Month Study Mode.*

- d. Open the single run controller and select start. Alternatively, you can open a saved SCT and click on the green run arrow at the top of the screen.

Ensemble or Mid-term Probabilistic Operations Model (MTOM) Mode

1. Receiving the CRMMS File Structure

Reclamation will officially provide the Ensemble file structure at least twice each year. The official releases are typically in conjunction with the January and April Colorado River Simulation System (CRSS) model runs.

The CRMMS model file will be included in the file structure and will be initialized with the Most Probable inflow forecast and default water use assumptions from the Most Probable 24-Month Study for that month, but will be in Ensemble (or MTOM) mode. Save the file structure provided by Reclamation on your computer and this will now become your CRMMS directory, where you can complete your CRMMS model runs and analysis.

***Note** – DO NOT modify the names of the **existing** sub-directories of the CRMMS file structure as this will break the model's environment variable linking structure that is required when running the model in Ensemble (MTOM) mode. It IS ok to add additional folders to this structure for analysis of model runs and to help prevent overwriting of output data.*

2. Setting Up the Environment Variable

Navigate to your CRMMS Directory and run the 1.SetCrmmsDirectory.bat script. The 1.SetCrmmsDirectory.bat script will create an environment variable on your computer called CRMMS_DIR that points to your CRMMS directory.

- a. For example, suppose your CRMMS directory is
C:\Modeling\CRMMS\MONTHYYYY\
- b. Run file - 1.SetCrmmsDirectory.bat from that location. Press any key to close the window after the variable has been set.
- c. The script has just set your environment variable, CRMMS_DIR, to your CRMMS directory (C:\Modeling\CRMMS\MONTHYYYY\)

Note – Setting the CRMMS_DIR environment variable is NOT required for running CRMMS in 24-Month Study Mode, but it IS required for running CRMMS in Ensemble (MTOM) mode.

3. Ensemble (aka MTOM) Mode Set-up

When setting up Ensemble mode, there are 3 things to consider: the inflow forecasts, output data folder and the run duration of the model.

1. The **Inflow Forecasts** folder includes two excel files containing the inflow ensembles.
 - a. CBRFC_EnsembleForecasts.xlsm
 - i. This includes a visualization tab for viewing all the traces for each input node
 - b. EnsembleForecasts.xls
 - i. This is used by the RiverWare model as the source of the inflows for each model run when performing the MRM run. **DO NOT** alter this spreadsheet.
2. There are two Output folders included in the package.
 - a. \$CRMMS_DIR/**Output Data** contains the output data in MS Excel format and .csv format.
 - i. CRMSS_EnsembleOutput.xlsm is a visualization app for analyzing ensemble data from an MRM run.
 - ii. EnsembleOutput.xlsx and CRMMScsvOutput.csv contain raw output data from RiverWare.
 - iii. CrmmsToCrss_Monthly.xlsx and CrmmsToCrss_Annual.xlsx are the initialization files for hybrid CRMMS Ensemble Mode-CRSS model runs
 - b. \$CRMMS_DIR/**rdfOutput** contains the output data in .rdf format
3. CRMMS Ensemble Mode (MTOM) has a run duration that spans 60 to 68 months, depending on the start month of the run. Model runs performed in February through September have a run duration that end in September of the last model run year (i.e. end of water year, similar to how run duration is set up in 24-Month Study Model, see Figure 1).

Note - To avoid overwriting output files provided with the CRMMS Ensemble Mode package when you perform your model runs, create a new sub-folder and copy all provided output files into that folder

4. Running CRMMS in Ensemble Mode

There are three ways to run CRMMS in Ensemble Mode:

1. Deterministic model run using the current loaded trace
2. Ensemble model run using all hydrologic traces
3. Running a single hydrologic trace using the MRM configuration

The following sections describe how to use CRMMS in Ensemble Mode to perform each type of analysis.

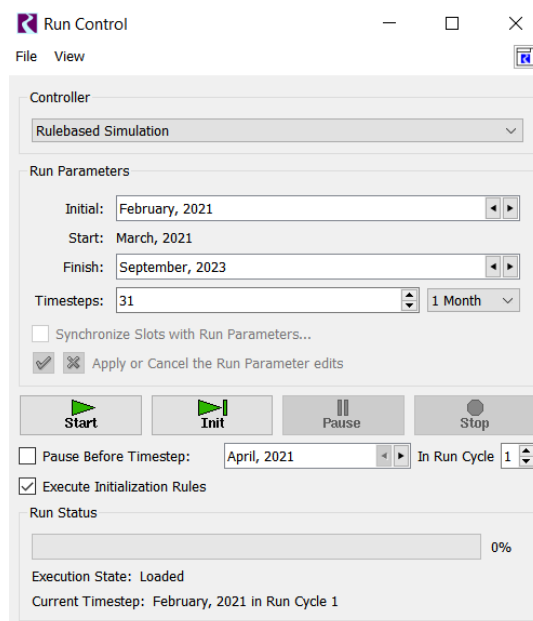
5. Running CRMMS in Ensemble Mode – Current Trace

The following instructions will result in a single trace being simulated with the hydrologic trace that is currently loaded in the model (ex., Most Probable inflow scenario, rule-based solving).

- a. Open the model (the ruleset will load automatically).
- b. If you are starting with the CRMMS model in Ensemble mode, no additional actions are required.
- c. If you are starting with the CRMMS model in 24-Month Study mode, go to the Scripts menu and select Script **5_Change Model to Ensemble Mode**.

Note – After running CRMMS in Ensemble mode it is recommend that you close the model and reopen the file if you are trying to return to 24-Month Study mode results. Shifting back to 24-Month Study mode after running the model in Ensemble mode will not repopulate the model with results from the official published 24-Month Study.

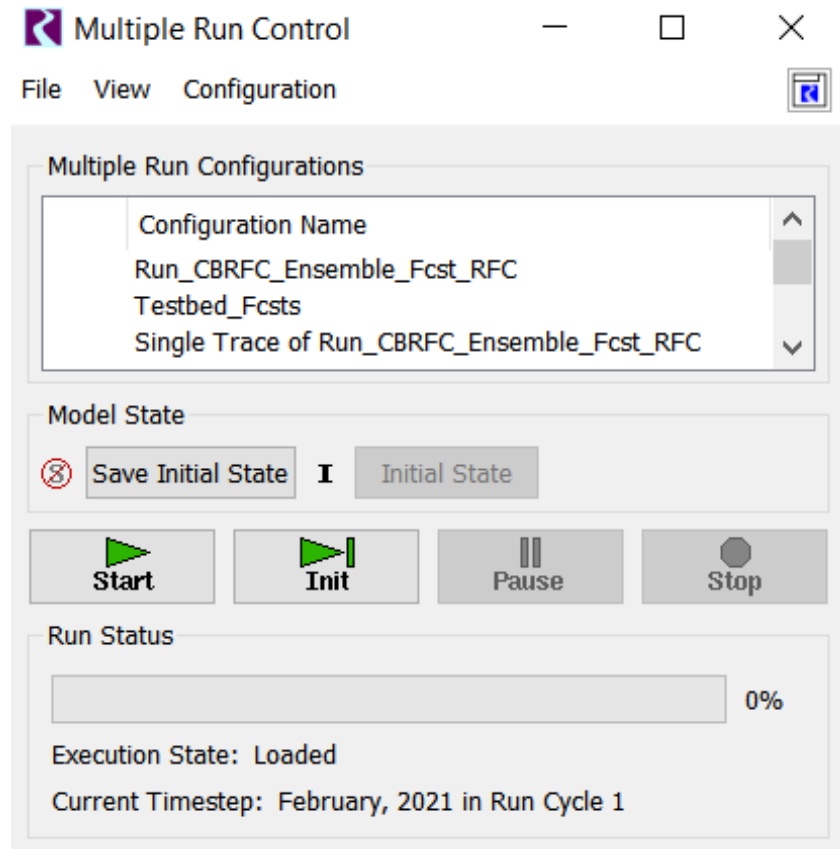
- d. Open the single run controller  and select start. Alternatively, you can open a saved SCT and click on the green run arrow at the top of the screen.



6. Running CRMMS in Ensemble Mode - MRM (All Traces)

The following instructions will result in an ensemble of traces being simulated. Follow instructions in previous section, *Running CRMMS in Ensemble Mode - Current Trace*, through item c before continuing below.

- a. Open the Multiple Run Control  and select the MRM configuration **Run_CBRFC_Ensemble_Fct_RFC**



- b. On the 'Output' tab, ensure the desired output formats are selected
- i. Default is to create MS Excel, rdf, and csv files

The screenshot shows the 'MRM Configuration - Run_CBRFC_Ensemble_Fcst_RFC' dialog box with the 'Output' tab selected. The 'Configuration' section at the top shows 'Name: Run_CBRFC_Ensemble_Fcst_RFC', 'Mode: Concurrent', 'Policy: Rules', and 'Input: Input DMIs'. The 'Output' tab contains several sections: 'Per Trace Output DMI' is set to 'Excel_Direct_OutputEnsemble'; 'RDF Options' includes 'Control File: \$CRMMS_DIR/Input Data/output.control', 'Data File' (empty), 'Timesteps: Must Match', and 'Allow Spaces In File Paths' (unchecked); 'Excel Options' includes 'Generate Excel Workbooks from RDF' (unchecked), 'Delete RDF Files' (unchecked), 'Slot Names: Index', and 'Configuration: Timesteps / Runs / Slots'; 'Generate Comma-Separated Values Files' is checked, showing a table with 'Name: CsvOutputMRM' and 'Number of Slots: 164'; and 'Generate NetCDF Files' is unchecked. Red arrows point to the 'Per Trace Output DMI' field, the 'Control File' field, and the 'Number of Slots' field. The bottom of the dialog has 'OK', 'Apply', 'Reset', and 'Cancel' buttons.

MRM Configuration - Run_CBRFC_Ensemble_Fcst_RFC

Configuration

Name: Run_CBRFC_Ensemble_Fcst_RFC

Mode: Concurrent

Policy: ☒ Rules

Input: ☒ Input DMIs ☐ None ☒ Traces ☐ Index Seq.

Description Output Run Parameters Policy Input Concurrent Runs

Per Trace Output DMI: Excel_Direct_OutputEnsemble

RDF Options

Control File: \$CRMMS_DIR/Input Data/output.control

Data File:

Timesteps: Must Match ☐ Allow Spaces In File Paths

Excel Options

☐ Generate Excel Workbooks from RDF ☐ Delete RDF Files

Slot Names: Index Rows / Columns / Worksheets: Configuration: Timesteps / Runs / Slots

☒ Generate Comma-Separated Values Files

Name	Number of Slots
CsvOutputMRM	164

☐ Generate NetCDF Files

OK Apply Reset Cancel

- c. On the 'Policy' tab, ensure the CRMMS policy set is selected.
- d. On the Input tab, the default configuration includes the Most Probable inflow trace, as well as the Probable Maximum and Probable Minimum traces if available. Including the three official forecasts in the MRM run means the First Trace will be set to 1 and the Number of Traces will be set to 38. If you want to exclude the three official forecast traces and only include the ESP ensemble, set the first trace to 4, and the Number of Traces to 35.

MRM Configuration - Run_CBRFC_Ensemble_Fcst_RFC

Configuration

Name:

Mode:

Policy: ☒ Rules

Input: ☒ Input DMIs

☐ None

☒ Traces

☐ Index Seq.

Description Output Run Parameters Policy Input Concurrent Runs

☐ Initialization DMI

Input DMIs

Repeat	DMI
38 x	Excel_Direct_InputEnsemble

☐ Append DMI

☐ Input Ensembles

Index Sequential / DMI Mode: ☒ Combinations ☐ Pairs

☐ Distributed Runs

First Trace:

Number Of Traces:

OK Apply Reset Cancel

- e. If any changes were made to the MRM configuration, Click OK and save the model with a new name.
- f. Select the MRM Configuration and click 'Start'.

7. Running CRMMS in Ensemble Mode - MRM (Single Trace)

The following instructions will result in a single trace from the ensemble being simulated (ex., Trace 15). Follow instructions in previous section, *Running CRMMS in Ensemble Mode - Current Trace*, through item c before continuing below.

- d. Open the Multiple Run Control and select the MRM configuration
Run_CBRFC_Ensemble_Fct_RFC
- e. On the 'Output' tab, you can deselect the output processes to avoid overwriting existing MRM output files
 - ii. Delete Excel_Direct_OutputEnsemble from the Per Trace Output DMI box
 - iii. Enter noOutput.controlfile.txt into the RDF Control File box
 - iv. Deselect the box Generate Comma-Separated Values Files

The screenshot shows the 'MRM Configuration - Run_CBRFC_Ensemble_Fcst_RFC' dialog box with the 'Output' tab selected. The 'Configuration' section at the top shows 'Name: Run_CBRFC_Ensemble_Fcst_RFC', 'Mode: Concurrent', 'Policy: Rules', and 'Input: Input DMIs'. The 'Output' tab contains several sections: 'Per Trace Output DMI' with a red arrow pointing to the empty text box; 'RDF Options' with 'Control File' set to '\$CRMMS_DIR/Input Data/noOutput.controlfile.txt' (indicated by a red arrow) and 'Data File' empty; 'Excel Options' with 'Generate Excel Workbooks from RDF' and 'Delete RDF Files' unchecked, 'Slot Names' set to 'Index', and 'Configuration' set to 'Timesteps / Runs / Slots'; and 'Generate Comma-Separated Values Files' (indicated by a red arrow) and 'Generate NetCDF Files' both unchecked. The bottom of the dialog has 'OK', 'Apply', 'Reset', and 'Cancel' buttons.

- f. On the 'Input' tab, change the number of runs to 1 and the First Trace to the trace number that you wish to run.

MRM Configuration - Run_CBRFC_Ensemble_Fcst_RFC

Configuration

Name: Run_CBRFC_Ensemble_Fcst_RFC

Mode: Concurrent

Policy: ☒ Rules

Input: ☒ Input DMIs

☐ None

☒ Traces

☐ Index Seq.

Description Output Run Parameters Policy Input Concurrent Runs

☐ Initialization DMI

Input DMIs

Repeat	DMI
38 x	Excel_Direct_InputEnsemble

☒ Append DMI

☐ Input Ensembles

Index Sequential / DMI Mode: ☒ Combinations ☐ Pairs

☐ Distributed Runs

OK Apply Reset Cancel

Traces

First Trace: 18

Number Of Traces: 1

Note - Traces 1 through 3 are the Probable Minimum, Probable Maximum, and Most Probable forecasts. To run ESP Trace 15 you must select First Trace = $15 + 3 = 18$.

- Click Ok, and then 'Start'.
- Note, that after this simulation finishes, you can re-run the same hydrologic trace again, by following the instructions in *Running CRMMS in Ensemble Mode - Current Trace*.
- When/if you save the model, make sure to Save As a new name, as the model will have data for the hydrologic trace recently analyzed and not as delivered.